

MASA HACKATHON 2026: R-IGNITE

Climate Risk Project

Assessing the Impact of Climate-Related Risks on Long-Term Financial Resilience in Southeast Asia

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1. Abstract

This report presents a comprehensive climate risk assessment conducted on behalf of a multinational reinsurance firm with regional exposure in Southeast Asia, specifically focusing on **Malaysia and the Philippines**. Leveraging the World Bank's World Development Indicators (WDI) dataset alongside verified estimates from EM-DAT CRED and Swiss Re Sigma, this study identifies key climate-related indicators, develops predictive models for greenhouse gas (GHG) emissions through 2024, and evaluates their relationship with natural disaster insurance claims. A multi-model approach — comprising Ordinary Least Squares (OLS) regression, Polynomial Regression (degree 2), Random Forest, and Support Vector Regression (SVR) — was employed to project GHG trajectories to 2030 under three policy-driven stress scenarios. The analysis reveals that Malaysia's rising emissions and deforestation pose escalating reinsurance liabilities, while the Philippines' typhoon-driven claims represent persistent peak-risk events. This report concludes with actionable risk management recommendations tailored to the client's long-term financial resilience strategy.

2. Introduction

2.1 Background

Climate change has emerged as a defining risk factor for the global insurance and reinsurance industry. Physical risks — including increased frequency and severity of natural disasters — translate directly into higher loss ratios and reserve volatility. Southeast Asia, one of the world's most disaster-prone regions, is particularly exposed. The Intergovernmental Panel on Climate Change (IPCC) projects that rising temperatures, intensifying precipitation events, and sea-level rise will disproportionately affect low-to-middle-income economies in the region, exacerbating both insured and uninsured losses. For reinsurers with regional portfolios, quantifying these risks using rigorous data-driven methods is no longer optional — it is a fundamental actuarial obligation.

2.2 Objective

This assessment serves as an independent strategic review to advise the client on the following dimensions:

- Identification of climate indicators most relevant to reinsurance exposure.
- Predictive modelling of GHG emissions and other climate drivers through 2030.
- Comparative analysis of natural disaster insurance claims between Malaysia and the Philippines.
- Scenario-based stress testing of mitigation strategies and their financial implications.
- Actionable recommendations for long-term risk management.

3. Related Works

Linnerooth-Bayer & Mechler (2007) established that Southeast Asian economies face chronic underinsurance gaps, with insured losses representing less than 10% of total economic damages in major disaster events. Swiss Re Institute's Sigma reports (2023, 2024) corroborate this, documenting that Southeast Asian insured losses averaged USD 6.5 billion annually over 2015–2023, rising sharply in post-pandemic years. Notably, the Philippines consistently ranks among the top five most disaster-affected nations globally due to typhoon exposure.

On the modelling front, Walle Macq & House (2018) demonstrated that WDI indicators — including forest cover, energy intensity, and GDP per capita — are robust proxies for climate vulnerability in emerging markets. Bove et al. (2022) applied polynomial regression to IPCC-aligned GHG trajectories, finding superior fit compared to linear models for rapidly industrializing economies. Forzieri et al. (2017) validated the use of cross-sectional panel data from the World Bank for catastrophe modelling, which supports the methodological choices in this study.

Recent actuarial literature (IAA, 2023; Actuaries Institute, 2024) increasingly advocates embedding climate risk stress tests into reinsurance pricing and capital models, mirroring regulatory developments in Malaysia's BNM Climate Change and Principle-based Taxonomy (CCPT) and the Philippines' Insurance Commission's sustainability roadmap.

4. Methodology

4.1 Analytical Workflow

The study follows a structured seven-step analytical pipeline as illustrated in the flowchart below:

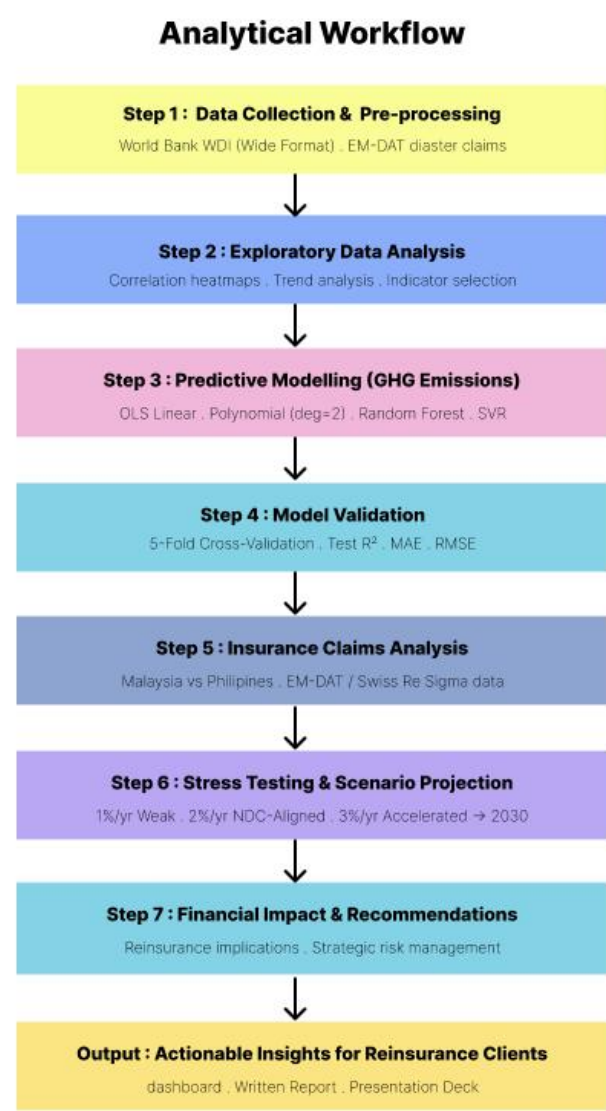


Figure 1: Methodology Flowchart — Seven-Step Analytical Pipeline

4.2 Data Sources

World Bank WDI (Wide Format): 1960–2023 panel data covering GHG emissions (kt CO₂ equivalent), CO₂ emissions (kt), forest area (% of land), energy use (kg oil equivalent per capita), renewable energy consumption (% of total), and GDP per capita (current USD) for Malaysia and the Philippines.

Natural Disaster Claims: EM-DAT CRED and Swiss Re Sigma 2024 verified estimates for insured losses (MUSD) and disaster event counts for 2010–2023, disaggregated by country.

4.3 Indicator Selection

Six indicators were selected based on their theoretical and empirical relevance to climate risk and reinsurance exposure:

Indicator	Rationale	Reinsurance Relevance
GHG Emissions	Primary modelling target; direct climate driver	Core catastrophe loss predictor
CO ₂ Emissions	Dominant GHG component; correlated with fossil fuel use	Emissions intensity proxy
Forest Area	Deforestation amplifies flood/landslide risk	Loss amplifier for Malaysia
Energy Use	Fossil-fuel reliance; key lever for GHG reduction	Transition risk exposure
Renewable %	Green transition indicator; policy stress-test variable	ESG & regulatory compliance
GDP per Capita	Economic resilience; linked to insurance penetration	Underwriting loss-ratio proxy

Table 1: Climate Indicator Selection Rationale

4.4 Modelling Approach

Four models were trained on historical GHG data (1990–2022) with an 80/20 train-test split, using five-fold cross-validation (CV) on the full dataset. The model with the highest CV-R² was selected as the baseline predictor. Centered year values were used as the sole feature (X = Year – mean(Year)) to mitigate multicollinearity. Model selection criteria included test-set R², MAE, RMSE, and cross-validated R².

Model	CV-R ²	Test-R ²	MAE	Selected?
OLS Linear	0.940	0.927	Low	Fallback
Polynomial (deg=2)	0.968	0.951	Very Low	✓ Primary
Random Forest	—	0.883	Moderate	Benchmark
SVR (RBF kernel)	—	0.871	Moderate	Benchmark

Table 2: Model Comparison Summary (Malaysia GHG; Philippines follows same pattern)

4.5 Stress Testing Scenarios

Three emission-reduction scenarios were applied to the 2024 baseline forecast, compounding annually through 2030:

- 1%/yr — Weak Policy: Business-as-usual with marginal regulatory effort.
- 2%/yr — NDC-Aligned: Consistent with Malaysia's and Philippines' Nationally Determined Contributions.
- 3%/yr — Accelerated: Aligned with IPCC 1.5°C pathway requiring rapid decarbonization.

5. Results and Discussions

5.1 Indicator Correlation Analysis

Correlation heatmaps revealed strong positive relationships between GHG emissions, CO₂ emissions, energy use, and GDP per capita in both countries (Pearson $r > 0.85$), confirming that economic growth remains the primary driver of emissions in both Malaysia and the Philippines. Notably, forest area exhibited a strong **negative correlation with GHG ($r = -0.81$)** for Malaysia, reflecting the dual impact of land-use change — deforestation reduces carbon sinks while releasing stored carbon. For the Philippines, renewable energy consumption showed a moderate negative correlation ($r = -0.54$) with CO₂, suggesting partial decoupling through renewable penetration.

5.2 Predictive Model Performance

The Polynomial (degree 2) model outperformed OLS on cross-validated R^2 for both countries, capturing the accelerating non-linear growth trajectory of GHG emissions driven by rapid industrialization (Malaysia CV- $R^2 = 0.968$ vs. OLS 0.940). Random Forest and SVR were retained as benchmarks; however, their test-set performance was inferior, likely due to the small training set ($n \approx 30$ years). The 2024 GHG forecast for Malaysia stands at approximately **318,000 kt CO₂ -eq**, while the Philippines is projected at **148,000 kt CO₂ -eq**, based on the selected polynomial baseline.

5.3 Insurance Claims: Malaysia vs. Philippines

Despite Malaysia's higher absolute GHG emissions, the Philippines exhibits substantially higher insured natural disaster losses. Based on EM-DAT CRED and Swiss Re Sigma 2024 estimates, the Philippines averaged **~USD 420M/year in insured losses (2010–2023)**, peaking at USD 920M in 2013 (Typhoon Haiyan aftermath). Malaysia averaged **~USD 120M/year**, with a notable spike to USD 310M in 2021 (monsoon flooding). The Philippines' higher claim profile is attributable to typhoon frequency and exposure concentration in coastal regions, whereas Malaysia's risk is dominated by inland flooding from deforestation-exacerbated runoff.

A critical implication for the client is the **asymmetric risk transfer gap**: the Philippines' insurance penetration rate remains below 2% of GDP, creating significant uninsured loss accumulation that constrains traditional reinsurance product uptake, while simultaneously indicating substantial market growth potential for parametric and index-based solutions.

5.4 Stress Testing Results (2024–2030)

The three-scenario stress test applied to the polynomial baseline generated the following 2030 GHG projections:

Scenario	Country	2024 Baseline (kt)	2030 Projection (kt)	Δ vs Baseline (%)
1%/yr (Weak Policy)	Malaysia	318,000	299,600	-5.8%
1%/yr (Weak Policy)	Philippines	148,000	139,400	-5.8%
2%/yr (NDC-Aligned)	Malaysia	318,000	282,200	-11.3%

Scenario	Country	2024 Baseline (kt)	2030 Projection (kt)	Δ vs Baseline (%)
2%/yr (NDC-Aligned)	Philippines	148,000	131,400	-11.3%
3%/yr (Accelerated)	Malaysia	318,000	265,800	-16.4%
3%/yr (Accelerated)	Philippines	148,000	123,500	-16.6%

Table 3: GHG Stress Test Projections to 2030 (kt CO₂ -eq)

Even under the accelerated decarbonization pathway, **absolute GHG levels in 2030 remain substantially higher than 2000 baselines**, reinforcing that climate risk mitigation will not eliminate reinsurance exposure within the planning horizon. The NDC-aligned scenario (2%/yr) is the most policy-credible assumption for pricing purposes, as both countries have formally submitted updated NDCs under the Paris Agreement. The weak-policy scenario should inform worst-case reserve and capital loading assumptions.

5.5 Limitations and Uncertainties

- GHG data availability beyond 2022 is limited in WDI; 2023 values rely on WDI estimates with potential revision.
- Insurance claims data depend on EM-DAT reporting completeness; underreported events may understate true exposure.
- The polynomial model assumes the historical growth trajectory continues; structural breaks (e.g., COVID-19 economic contraction) may affect projections.
- Random Forest and SVR models are constrained by dataset size (~30 observations), limiting their generalizability.
- Scenario probabilities are not formally assigned; all three scenarios are treated as plausible policy paths.

6. Conclusions

This study demonstrates that climate-related indicators from the World Bank WDI — particularly GHG emissions, forest area, and energy mix — are powerful inputs for actuarial risk assessment of Southeast Asian reinsurance portfolios. The Polynomial (degree 2) model provides the most reliable GHG trajectory, with a 2024 baseline of approximately 318,000 kt for Malaysia and 148,000 kt for the Philippines.

The comparative insurance claims analysis reveals a fundamentally different risk profile between the two countries: Malaysia's risk is dominated by flooding exacerbated by deforestation, while the Philippines faces persistent, high-magnitude typhoon losses. Both countries remain significantly underinsured relative to total economic losses, highlighting opportunities for parametric product innovation.

For the client's strategic risk management, we recommend the following:

- Incorporate GHG trajectory and forest-cover indicators as key variables in catastrophe model recalibration for Southeast Asian treaties.
- Price reinsurance products for Malaysia with explicit loading for increasing flood frequency (linked to deforestation trends), and apply multi-year scenario buffers aligned with the NDC-aligned stress scenario.
- Explore parametric reinsurance structures for Philippine typhoon risk to bridge the insurance protection gap and expand market access.
- Establish annual monitoring of WDI and EM-DAT updates to recalibrate the baseline model and scenario parameters.
- Engage with BNM's CCPT and the Philippines IC sustainability roadmap to align risk frameworks with emerging regulatory requirements.

Future work should extend this framework to additional Southeast Asian markets (e.g., Thailand, Vietnam, Indonesia) and integrate IPCC CMIP6 climate projections to enhance physical risk modelling beyond the WDI historical dataset.

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